A PROJECT REPORT ON

BEHAVIOR-BASED PERSONALIZED ADVERTISEMENT RECOMMENDATION

Msc Data Science

University of Europe for Applied Sciences.



Instructors

Mr. Ebby George Varghese

Dr. Humera Noor Minhas

Presented By

DEVI VARA PRASAD CHINTAPALLI - 33023056

PRAJWAL GURIVINDAPALLI - 30630894

Contents

1. Introduction & Problem Motivation....4
1.1 Project Overview....4
1.2 Problem Motivation....4
1.3 Motivation for Automation....5
1.4 Process Diagram....6
2. Problem Statement....7
3. Background and Literature Review....7
4. Data....10
4.1 Data structure....10
4.2 Feature Description....10
4.3 Nature of the Dataset....10
5. Data Preprocessing....11
5.1 Data Cleaning....11
5.2 The Categorical Variable as a Healthcare Data Element. 12
5.3 Feature Transformation 12
5.4 Train-Test Split....13
6. Exploratory Data Analysis (EDA) 13
6.1 Click-Through rate (CTR) Analysis. 13
6.2 The analysis of device-based engagement 14
6.3 Banner Position Analysis....14
7. Graphical and Insightful Data....15
7.1 The advertisement category visualization of CTR....15
7.2 visualization CTR by Type of Devices....16
7.3 Use of Visualization to make decisions....16
8. Highlights of Significant EDA Results. 16
9. Methodology....16
9.1 Problem Formulation....17
9.2 Feature Selection 17
9.3 Model Selection....18
9.4 Model Training....18
9.5 ModelEvaluation....18
10. Tools and Technologies....18
11. Results and Discussion....20
11.1 Exploratory Insights....20
11.2 Model Performance....20

11.3 Business Implications……21
12. Limitations……22
13. Future Scope……22
14. Conclusion……23
15. Reference……23

1.Introduction & Problem Motivation

1.1 Project Overview

The online advertising business has experienced an impressive expansion due to the immense utilization of online-connected devices, social media, and online shopping applications. Though virtual platforms provide advertisers with access to enormous data about users, many advertising systems continue to have difficulties in displaying appropriate ads to users. This usually implies that the users are subjected to advertisements that are not in their interest, web surfing, or intentions. This in turn results in reduced participation, ad nausea and poor user experience. This leaves the advertisers with poor targeting strategies, lack of target audience, poor return on investment (ROI), and poor click-through rates (CTR).

To prevent these problems, this project proposes a Personalized Advertisement Recommendation System (Behavior-Based) that utilizes the methods of machine learning and data science. The system examines the history of user behavior, the surrounding information, device capabilities, as well a.s advertisement characteristics to determine the likelihood of a user clicking a specific ad. Representing the user click behavior in a supervised learning method is through the target variable. The project includes predictive modeling, engagement pattern visualization, feature engineering, and exploration data analysis in addition to comprehensive data preprocessing.

This study findings indicate that such behavioral and contextual factors as the type of ads, as well as the type of device or ad position, influence user attention to ads significantly. It is possible to enhance the user engagement rates and maximize advertising efficacy with the proposed system implying the utilization of these insights to propose more relevant ads. The project provides customizable architecture of advertising systems that can be extended and proved to be useful in addressing meaningful digital marketing challenges.

1.2 Problem Motivation

The project was an initiative of the practical issues in the industry and the wish to get to know more about the system of personalization. Today, individuals are overwhelmed with

advertisements on the Internet and social media, websites, and applications. The number of ads, therefore, makes people not pay as much attention and feel irritated.

When users are presented with irrelevant ads, it becomes frustrating to navigate it and the site appears to be subpar. Individuals desire to watch something that suits their needs and interests. Failure to do so may make them quit using the platform.

On the part of the advertiser, digital advertising is very expensive. Without targeting the ads, the money will go down the drain and people will not even get involved or make any purchases. Relevancy of ads is the major way to reap the maximum out of your investments and to be ahead of the competition.

It is an opportunity to apply what you have studied about data science such as checking out data, manipulating features, applying supervised learning, and determining the goodness of a model in the real world. A good illustration of how making decisions based on the use of data can benefit a business is in the recommendation of ads.

1.3 Motivation for Automation

The more I use the internet, the more these systems learn about my preferences and interests. They try to understand what I like by analysing my past behaviour and predicting which advertisements are most likely to catch my attention. Factors such as the time of day, the device I use, and the websites I visit play an important role in this process.

These systems help advertisers by showing ads to users who are genuinely interested in them. This leads to higher click-through rates and better sales outcomes. Advertisers also benefit by using their advertising budget more efficiently instead of targeting uninterested users. At the same time, users have a better experience because they are shown relevant and meaningful advertisements rather than random ones.

Overall, personalized advertising systems create a win-win situation. Users receive useful and interesting ads, while advertisers achieve better results. The main goal is to deliver the right product to the right person at the right time.

In this project, I focus on developing and deploying a machine learning–based ad recommendation system using user behaviour data. The objective is to predict whether a user is likely to click on an advertisement based on behavioural, contextual, and ad-related

features. Through analysis and system design, this project aims to model and forecast user behaviour on an advertising platform.

1.4 Process Diagram

The following diagram illustrates the high-level process flow of the "Behavior-Based Personalized-Advertisement-Recommendation"

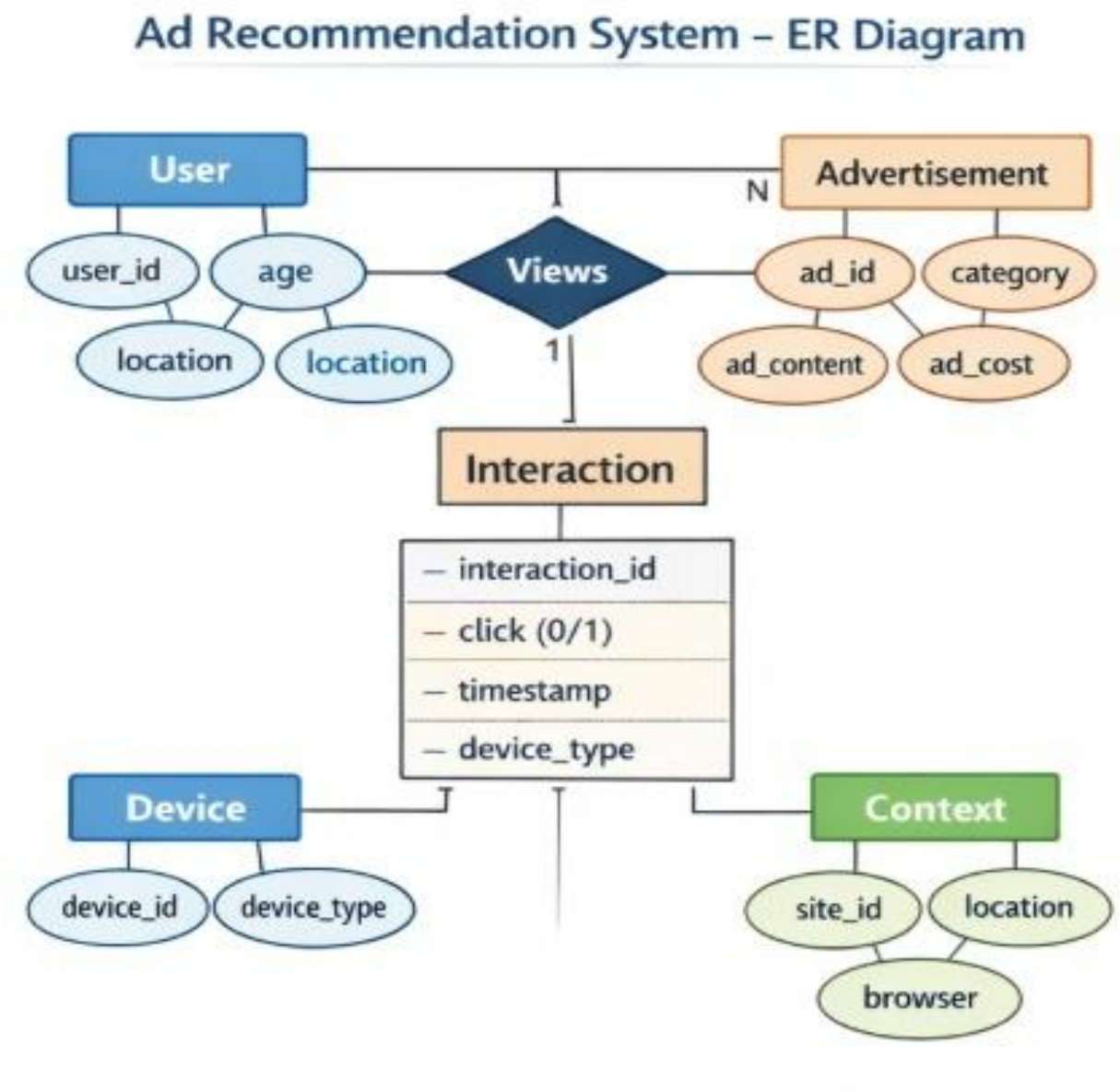


Figure-1.1: Flowchart of the system

While the Advertisement entity keeps ad data like category and cost, the User entity keeps user details like age and location.
Every time a user views an advertisement, it is documented as an interaction.

The device type, timestamp, and click behaviour are recorded by the Interaction entity.
Device type, browser, and site location are among the extra details provided by Device and Context entities.
In general, it clarifies the data-level relationship between user behaviour and context and advertisements.

2.Problem Statement

Online advertising has emerged as one of the major sources of income of websites, mobile apps, and online platforms in the existing digital ecosystem. As users continue to do their business online, they are constantly bombarded with many advertisements in the form of display advertisements on sites, in-app and social media advertisements. Nonetheless, much of these advertisements are not applicable to individual users since they are not consistent with the interests of the user, his or her browsing behavior and contextual preference. The effect of this excessive exposure to the irrelevant ads is ad fatigue, less engagement with the users, and the bad browsing experience.

On the part of the advertisers, the poor targeting of the advertisements may result in poor click-throughs, poor advertising budget utilization, and low returns on the investment. Conventional systems of distribution of advertisements tend to use rigid rules, simple demographic data, or broadly based assumptions, which are not dynamic when it comes to the dynamism of user actions. Despite the fact that there is a vast amount of data in user interaction every day, several existing systems fail to take advantage of such data to enhance personalization.

The problem is that it is difficult to analyze historical data of user interaction to be able to model the user behavior and work with anonymized, high-dimensional, and skewed data. It is a complicated issue as it is always difficult to predict whether a user will pay attention to a certain advertisement or not because of various contextual and behavioral circumstances. Consequently, it is necessary to have a smart and data-oriented system that can anticipate user attention and suggest the appropriate advertisements. The proposed project will provide some solutions to this issue and develop a behavior-based, personalized advertisement recommendation system that utilizes machine learning tools to enhance the relevance of the advertisements, the user experience, and the effectiveness of advertisements.

3.Background and Literature Review

During the past decade, customized recommendation systems have gained extreme popularity in research communities and in real life businesses. Recommendation systems attempt to provide users with something they think they will enjoy based on what they like,

what they do, and what state they are in at a particular moment, such as when you are shopping on the Internet, watching videos or on social media. However, they are difficult to use in online advertisements due to the lack of data, the issue of privacy, and the ability of the people to decide and change their mind rapidly.

Initially, the online advertisements were primarily targeted at a group of individuals based on various factors such as age, gender, and geographical locations. The point was that when individuals were in a common group, they were likely to prefer the same things. It was an easy method of directing advertisements, yet it did not really have a chance to know what the individual desires. The advertisements were usually very general and not very interesting hence people did not give much attention to them.

However, along with the increased amount of data companies gather, they began to target ads on what people do on the Internet. Advertisers can make guesses about things people are interested in by observing things such as what sites they visit, what they search in search engines, and how they engage with the content. It has been shown that this method of targeting advertisements results in many more clicks than simply applying simple information such as age. Once advertisers focus on what people do rather than only on who they are they are able to determine what people want in a far much better way.

One of the main challenges that has received a lot of research in online advertising is the ability to predict the click-through rate (CTR). The reason why logistic regression has been applied many times is that it is a simple and easy to comprehend method, which is able to work with numerous data. Although it is simple, it can work when the correct features are used. However, as the data on advertising is becoming increasingly complex, individuals are experimenting with superior models.

Random Forest and Gradient Boosting Machines, decision trees and ensemble approaches are popular in predicting CTR. They can arrive at complex relationships and interactions among the various features. These models can know intricate things automatically, such as the relationship between the device type, ad-location, and user context. Research indicates that the ensemble models tend to perform better when compared to the linear ones in prediction of things when the data is of various types.

Advert systems now make use of deep learning. Neural networks such as deep feed forward networks are useful in working with large quantities of various kinds of data. Deep learning models learn various trends, which makes them guess the trivialities of how individuals behave. These models require a lot of data and computing resources thus may not be suitable for everyone.

The locality of the advertisements is important. The device you are using, the time, your internet and where you see the ad on the page are all things that modify the response people have towards adverts. Phone advertisements attract more attention as compared to computer adverts. Advertisements belonging to the first page are likely to be more clicked.

Online ads are concerned with privacy. Most datasets conceal the confidential data by transforming user and campaign information into categories. This ensures that the users are safe, yet this complicates understanding. Surprisingly, despite the concealed information, there is a lot to guess if you are aware of what you are doing.

This project takes the known and integrations of how people move, the scenario, and the nature of ads into a single method of making a guess as to what transpires. It is not theory, it is really seeing data and making pictures to be able to come up with ideas we can use. The system is effective in the real world since we are interested in the results and dig in the data nevertheless, it is not less grounded in the research.



Figure-3: Background process

4.Dataset

This project applies to a synthetic dataset that is based on the Avizo Click-Through Rate (CTR) dataset. The dataset is usual in both scholarly and industrial investigations and attempts to simulate user-ad engagements of online advertisement as every line of data represents a single advertisement shown to a client.

4.1 Dataset Structure

This table contains 1,000 rows where the data is shown on what users do, the type of ads, and other information. It is primarily to determine the probability of a person clicking an advertisement.

4.2 Feature Description

• **click:** The dependent variable on whether the user who saw the ad was able to click on it or not. 1 and 0 will indicate a click and no click respectively.

• **site category:** Represents the type of the site in which the advertisement was placed, i.e., technology, sports, fashion, food, or gaming.

• **device type:** The device can be a mobile, desktop, tablet or any other device used by the user.

• **device_conn_type:** Defines the type of network connection, e.g. Wi-Fi or mobile data.

• **bannerols:** Determines where the advert will be situated on the webpage, which determines the visibility and interaction.

• **hour:** Indicates the time when the impression of an advertisement was received. Time-based capabilities assist in capturing time-related behavior of users.

• **C14–C21:** These are anonymous categorical and numerical attributes which reflect advertisement, campaign, and contextual characteristics. They do not reveal their meanings but have high predictive power and they have regular applications in CTR prediction exercises.

4.3 Nature of the Dataset

This data is constructed to be like the real-world advertising scenarios but the data about users remains confidential. We have ensured that we do not show any sensitive information and so it is suitable in school research. Although it is unnatural, it bears beneficial patterns

and variations which are required to gain knowledge about the information and draw conclusions. The target variant distribution involves the distribution of the target variable.

The variable of clicks is discreet, i.e. there is imbalance in classes. It is typical of ad data, in which non-clicks greatly outnumber clicks. This complicates training and verification of our models, and it is necessary to choose appropriate metrics to evaluate them.

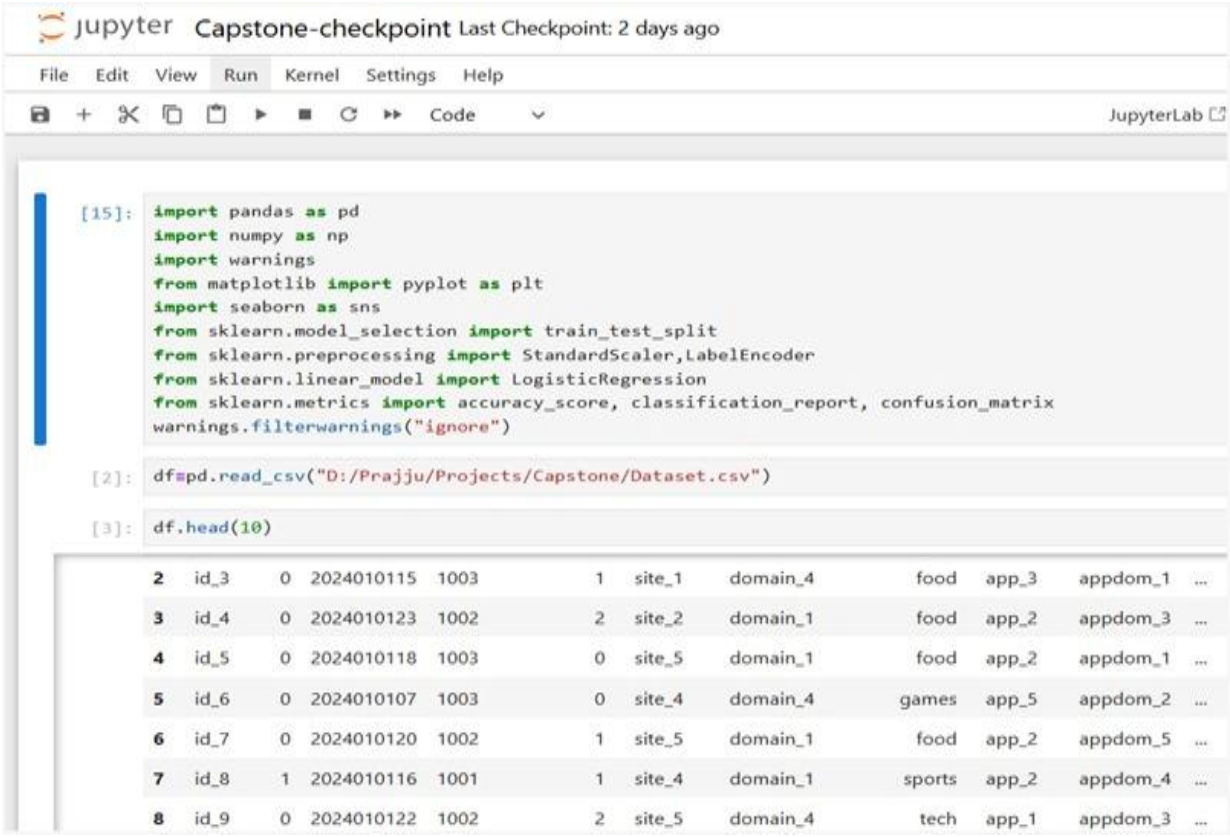


Figure 4.3: Dataset

5.Data Preprocessing

It is of utmost importance to prepare your data in data science. In case of bad info, your machine learning would not be effective or reliable. Online advertisements harvest information across a myriad of locations, and this fact renders things to be untidy with mistakes or absent elements. You need to clean and prepare the data then you can begin to look at it and construct models.

5.1 Data Cleaning

First, I verified the information regarding missing and duplicate information and errors. The data was not generated naturally and thus there were no missing data or duplications. Nevertheless, I verified the information to ensure that everything was correct and

reproducible.

I checked every value to ensure that the values were reasonable. As an illustration, I verified the machine and network type to ascertain that it corresponded to the available accepted categories. This is done to ensure that the analysis is not ruined by any misguided or unfamiliar values.

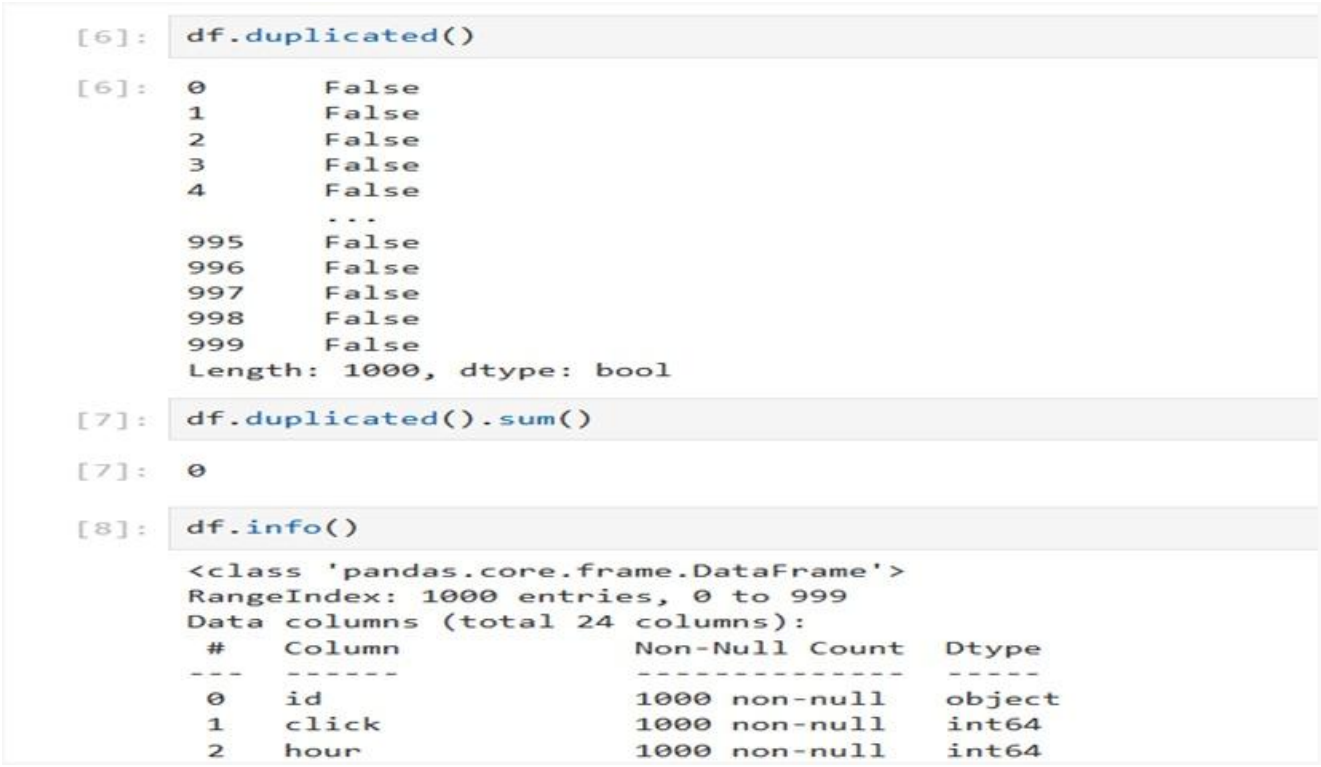


Figure 5.1: Data Processing

5.2 The Categorical Variable as a Healthcare Data Element

The data is highly categorical, with such variables as ad category, device type, banner position, or anonymous campaign IDs. The categories were transformed into numerical data because machine learning algorithms normally require numbers.

To make it easy to figure out, the categories were retained as they were on the first analysis and visuals. In the case of predictive models, these encoding schemes such as label encoding or one-hot encoding were applied, depending upon the algorithm. It was aimed at retaining significant connections among categories, keeping the dimensions small.

5.3 Feature Transformation

Other things that are named with numbers, such as the hidden stuff named C14 to C21 are measured differently. In the event we wish to employ math tricks that are concerned with the

size of these numbers, we must be careful to put them all on equal footing. Although not all methods are so concerned with this, once the numbers are brought to the same scale, we can compare them better. Besides, it also allows us to use other kinds of math in the future.
We also borrowed time-based information to find out how others behave at various times of the day. A glance at the hour will enable us to determine when people are the most active. This can dictate the time of placing ads.

5.4 Train-Test Split

We typically divide the data into two sets, one of which is a training set and the other is a testing set to measure the performance of a model. We learn the model based on the training set, the training set consists of the past data. This is followed by checking whether the model can give good guesses on the testing set, that is data which it has never seen. In that manner, we will be more certain that our model is doing the job properly and not merely learning the training data by heart.

6. Exploratory Data Analysis (EDA)

It is extremely essential to check out the data to get what is truly happening. It assists us in identifying some patterns, identifying the relationship between things and identifying the weirdness that may disarrange things in future when we are developing our model. In this project, I primarily considered the behavior of the users in response to the type of ad, the device they are using as well as some other information.
The analysis involves the advertisement category as a form of advertisement that an organization employs to reach various customer groups via mass media channels.
We have used bar charts to determine how many advertisements fell under various categories. We have checked such categories as tech, sports, fashion, food, and gaming to see which ones appeared the most.
The findings indicated that some types of ads are considerably more prevalent. This is understandable as that is the case with ads online. It is good to know what types of categories contain a plethora of advertisements since they can bias things unless we take care.

6.1 Click-Through rate (CTR) Analysis

One of the most important metrics of the interaction between people with online ads is the click-through rate. It is determined by taking the number of ad clicks and dividing it by the number of times the ad had been displayed on a particular set of features. In the present project, CTR was studied in relation to various forms of ads and types of devices.

We have analyzed that there are types of ads that are more likely to be engaged with as compared to other types. This implies that individuals have more interests in some kind of content. As an illustration, technology advertisements may attract more clicks as individuals tend to use technology.

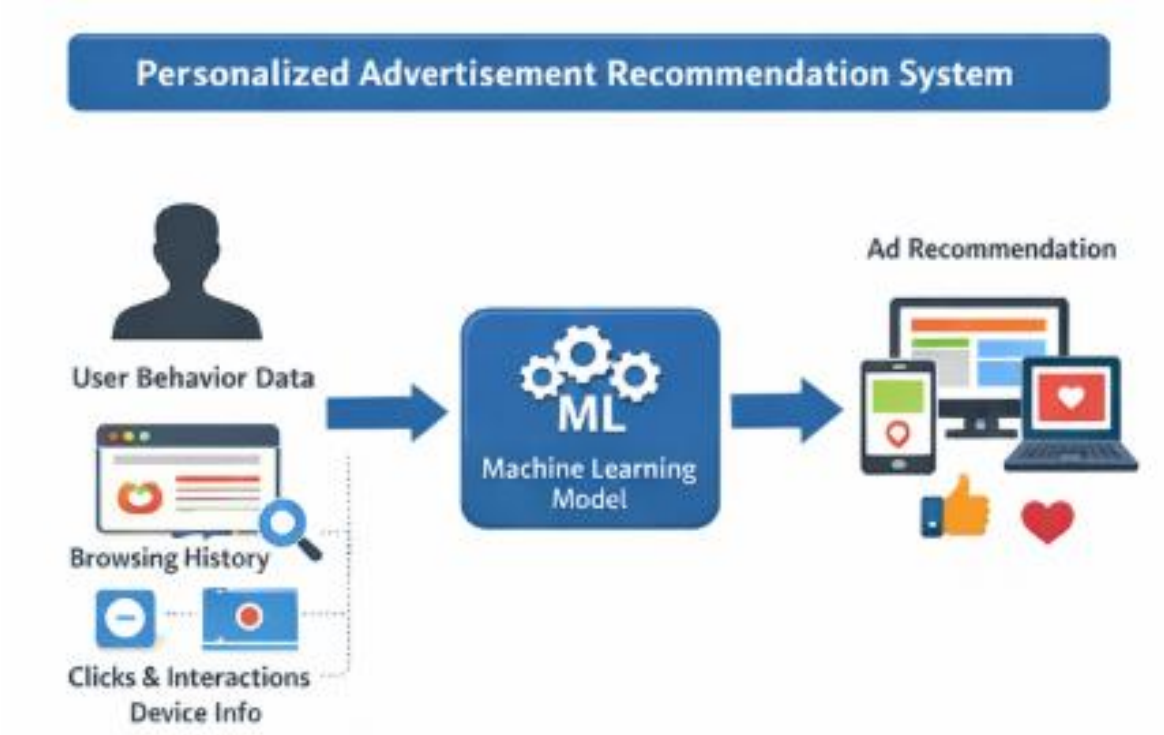


Figure 6.1: ad clicks and dividing

6.2 The analysis of device-based engagement will be conducted as part of the section

The interaction of people with things online is altered by the way they use various devices. We contrasted mobile phone and desktops and discovered that mobile users do more clicks on advertisements.

This is logical as individuals tend to use phones and have a quick look at something, which may result in additional clicks. Therefore, it is necessary to ensure that adverts are attractive on phones.

6.3 Banner Position Analysis

The position of an advertisement on a webpage affects the number of people that view the ad. Advertisements seen directly receive more attention. We compared the click-through of ads in various positions and realized the position of an ad has a huge impact on the responsiveness of

users. Our data revealed that the position of an ad can be central or visible, and it can result in more clicks compared to those that are not visible.

7. Graphical and Insightful Data

Visualization is one of the best methods of sharing what you have learned in data analysis. In relation to this project, we made use of various charts to demonstrate the involvement of the users and support our analysis.

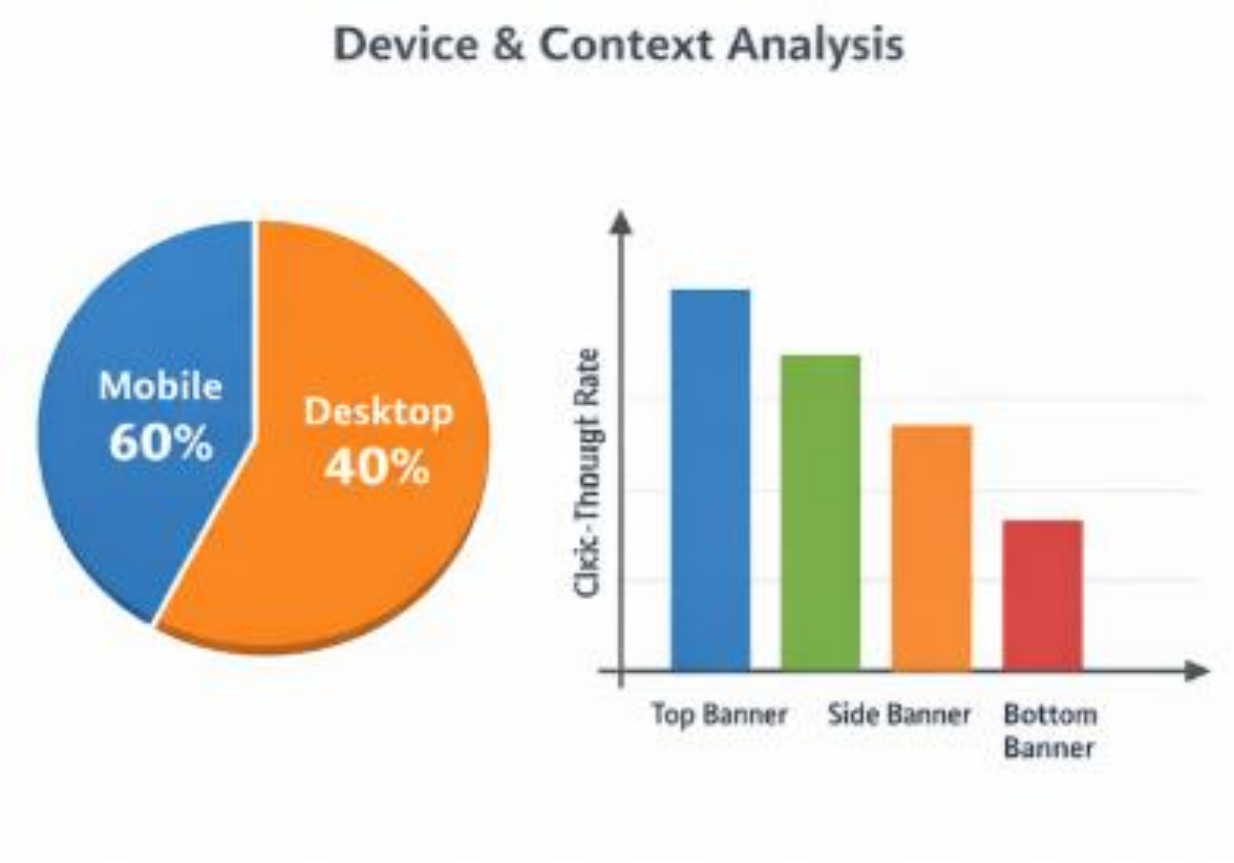


Figure 7: charts to demonstrate

7.1 The advertisement category visualization of CTR

We presented a horizontal bar chart to indicate the rate of click throughs of the various types of ads. With the help of this chart, it is not difficult to compare the effectiveness of each category in capturing attention and highlighting the best ones.

Analytical Insight: As the chart indicates, there are certain forms of ads which are always performing better. This is to say that what is being talked of in the ad influences the clicking of people. This information can be used by the advertisers to target the most effective ad types.

7.2 visualization CTR by Type of Devices

The bar chart represents the device-based click-through rates (CTR). People seem to have a greater number of clicks on phone ads.
Analytical Insight:
This implies that the device that a person is on is important. To enhance targeting, the ad systems must make use of device type.

7.3 Use of Visualization to make decisions

Raw data is transformed into helpful information using visual analytics. People's graphics make understand user behavior and make smarter decisions by displaying complicated relationships in straightforward ways. These insights are indeed significant with regards to advertising since they can make ad campaigns more effective and monitor the results of the ad campaigns.

8. Highlights of Significant EDA Results

The exploration analysis performed in the given project has shown several significant observations:

1. The level of user engagement in different advertising types differs greatly.

2. The kind of device used is a major factor in click-through.

3. Placing ads influences views and interactivity.

4. Behavioral and contextual characteristics can be a useful attribute to personalization.

These results confirm the significance of considering different types of features into the predictive model and the usefulness of behavior-based advertisement recommendation systems.

9.Methodology

We constructed this recommendation system of ad using a traditional approach of data science that prioritizes user behavior. It is teaching a computer to learn based on past user data to make an educated guess on what type of ads they will appreciate in future.

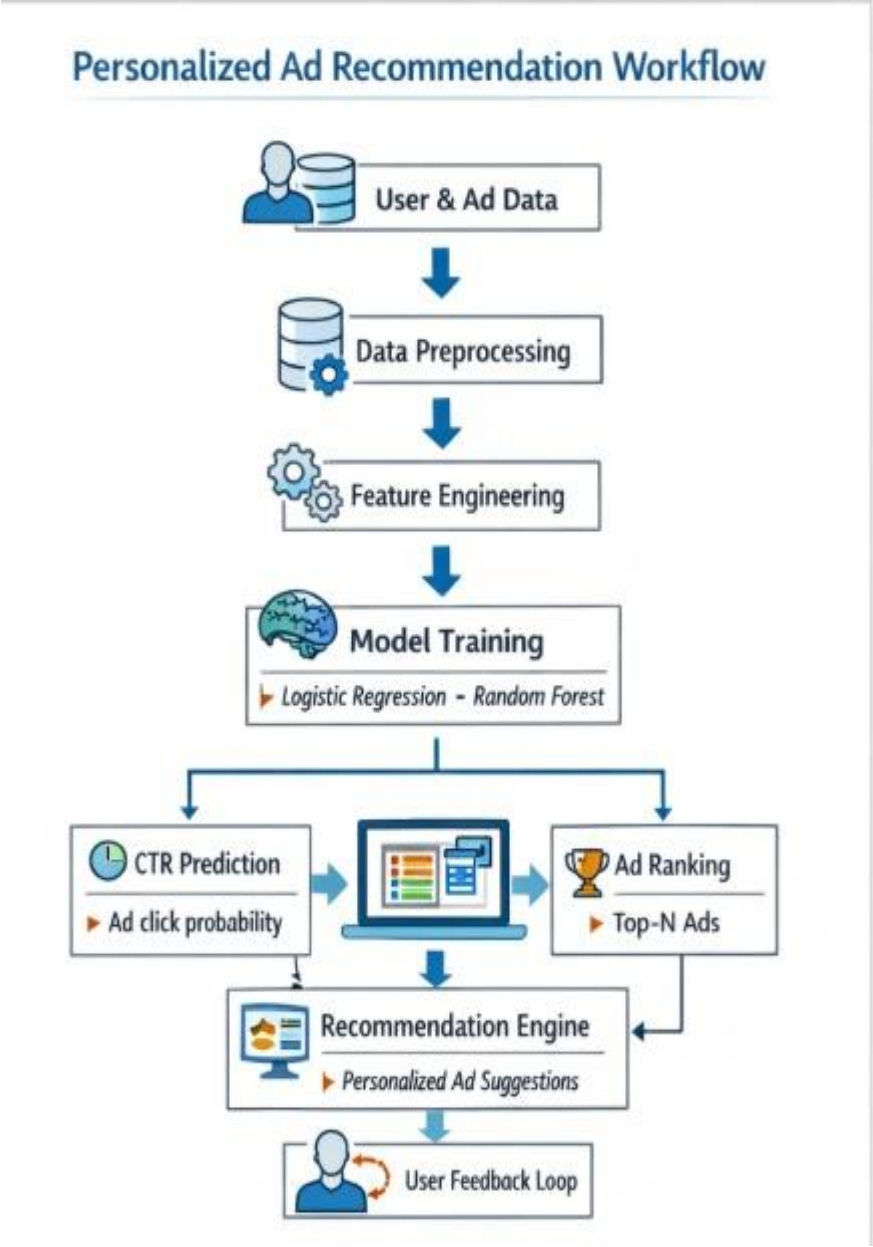


Figure 9: workflow

9.1 Problem Formulation

We take the recommendations of ads as a yes/no question: Will a user just click on this ad? If they click, it's a 1; if not, it's a 0. This is the way most predictions based on click-through rates are performed.

9.2 Feature Selection

It is important to select the right features as getting the wrong features may confuse the effective performance of the model. We chose the following features due to data observation

and prior knowledge:

- Behavioral Characteristics: Patterns of historical behavior of clicking on the dataset.
- Contextual Features: Type of device connected to, type of network connection and time of impression.

Aspects in advertisement: Type of advertisement, location of the banners and the anonymized campaign.

C14-C21: Anonymized Features- Coded variables containing the characteristics of the ad and campaign.

All these properties describe the context of interaction and relevance of the ads to the user.

9.3 Model Selection

There are several supervised learning techniques which are effective in the prediction of click-through rate (CTR). In this project, we are considering both predictive as well as understandable models. We shall begin with Logistic Regression as it is easy and understandable. Decision Trees and Random Forests can also be used. They are efficient in identifying complicated patterns and interactions between various features.

9.4 Model Training

Our mode of thinking was trained on a set of data, and it is learning to identify patterns that can correlate what is put in with the likelihood of a person clicking. To ensure that we did not over-train the data we adjusted the settings so that it could perform optimally.

9.5 Model Evaluation

To find out the working efficiency of the model, we applied typical tests, such as accuracy, precision, recall, F1-score, and ROC-AUC. Such tests provide us with the complete picture regarding the predictive power of the model, particularly in cases of uneven data when simply observing accuracy is misleading.

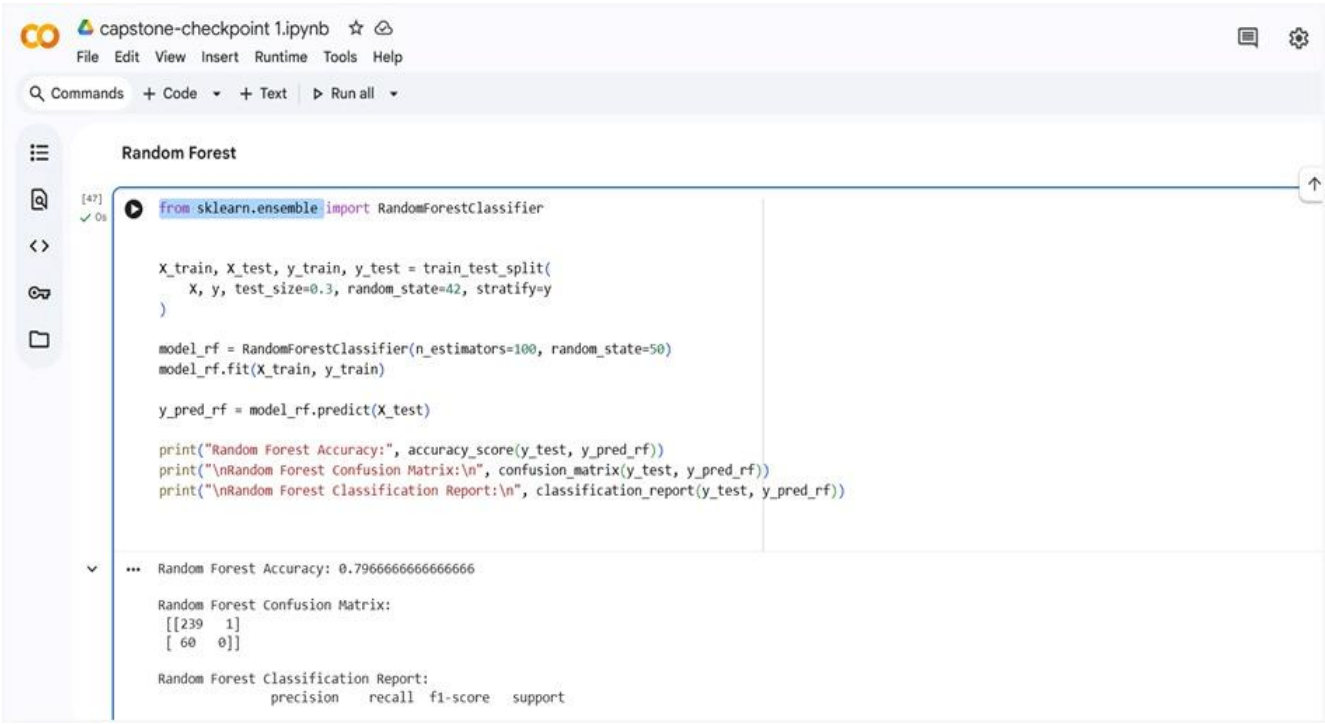
10. Tools and Technologies

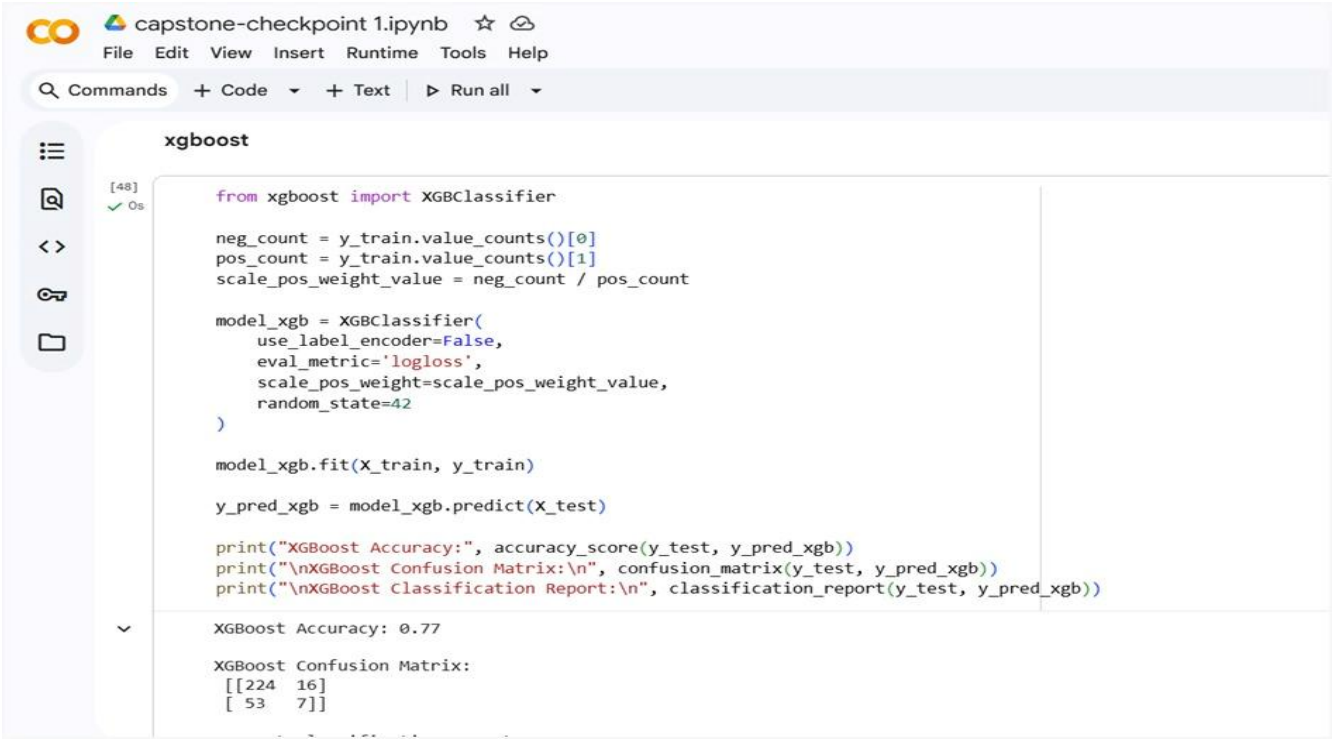
The project relies on the combination of programming tools and libraries that are popular in the field of data science and machine learning.

Programming Language: Python.

- Data Manipulation: Pandas.
- Data Visualization: Seaborn.

Machine Learning: Scikit-learning.
Knowledge: Jupiter Notebook Development Environment.
The tools offer a dynamic and effective platform for analyzing data, visualizing and developing models.





Figures 10: programming tools

11. Results and Discussion

The findings of this project demonstrate the success of the use of behavior to provide personalization in the online advertisement system.

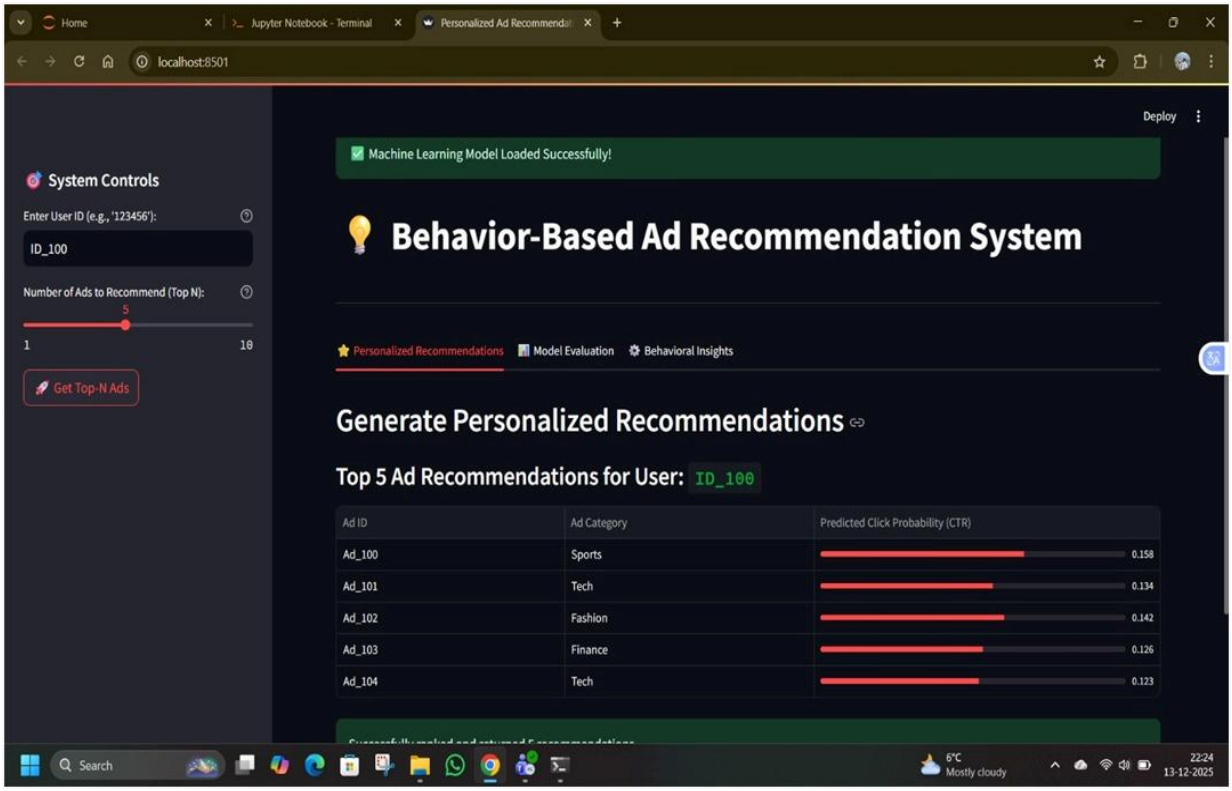


Figure 11: ads more as compared to computer users.

11.1 Exploratory Insights

Our data analysis indicates that there are certain apparent patterns in the way individuals engage with advertisements. We observed that there are certain categories which received much more clicks and this informs us that people are very interested in those. We also observed that people on phones and tablets tend to click on ads more as compared to computer users.

11.2 Model Performance

The prediction model has determined the relationship between user behavior, behavior and context, and ad features. Checks indicate that it is less arbitrary than guesses and it is picking up on actual engagement patterns.

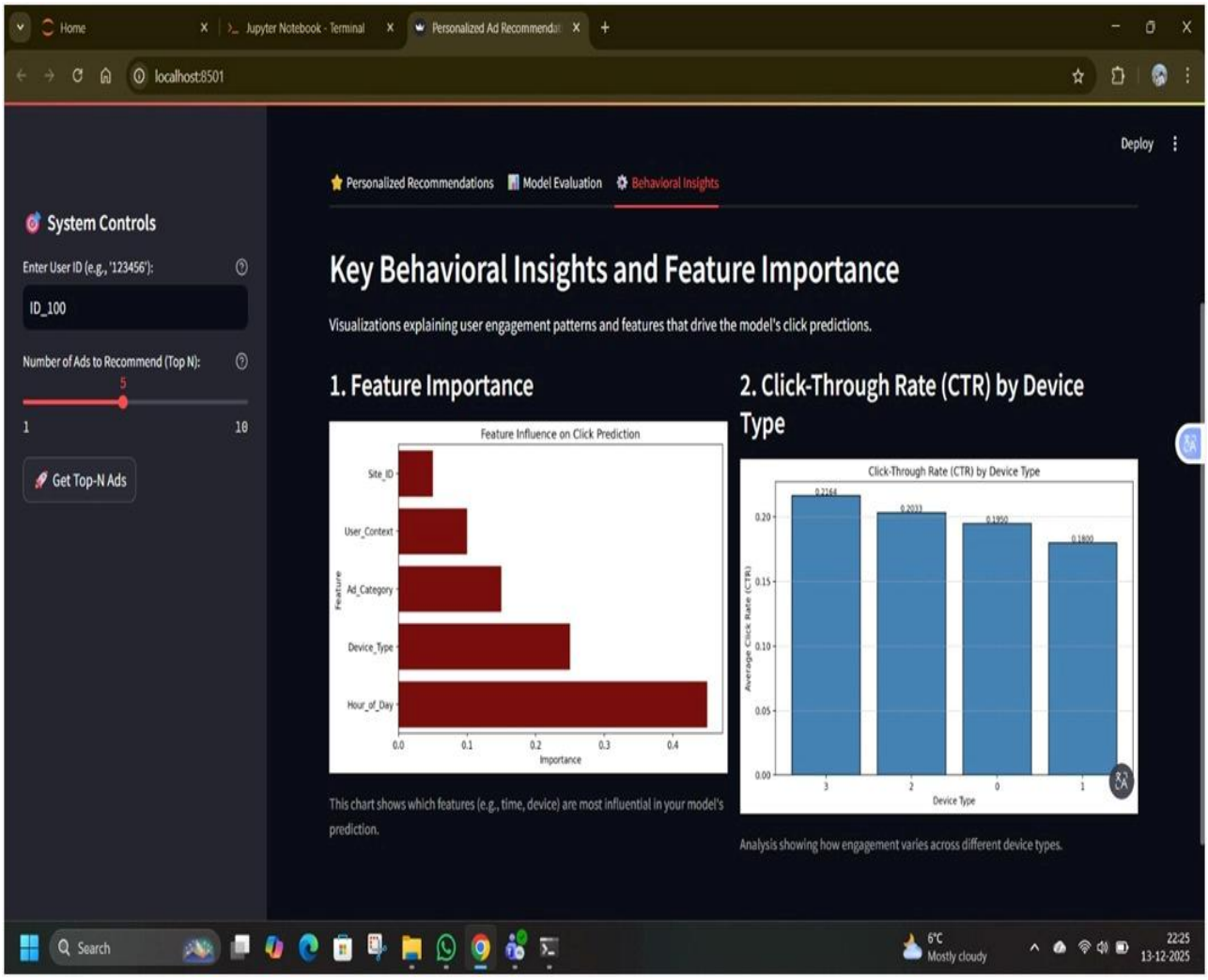


Figure 11.2: engagement patterns.

After creating the Layers, a relation must be made between images and users to find a group of users that could be the potential customers for an advertising image. This relation is created by connecting the two defined graphs and building a multi-layer graph. The common elements to link the layers are the keywords; consequently, the interconnection of layers is based on the common keywords among both layers. Afterward, the potential customers for each image are identified with in the set of users.

11.3 Business Implications

In business terms, these findings demonstrate that targeted advertising can make individuals interested and facilitate maximization of advertising budgets. Advertisers will be able to maximize their payoff by focusing on the categories and devices performing well and making their campaigns more successful.

12. Limitations

Although this project is contributory, it has several limitations:

1. **Synthetic Dataset:** The data utilized is artificial and might be unable to include all complexities of real data of advertising.

2. **Limited Data Volume:** Its size is not very large compared to industry level datasets.

3. **Static Behavior Modeling:** The model fails to take into consideration real-time alterations in user behavior.

4. Privacy Constraints: Anonymized features are less interpretable and explainable.

13. Future Scope

The potential for further improvement of this project is enormous:
In the future, we will run the algorithm on more datasets and improve its efficiency to make it more suitable for large datasets. Furthermore, we will incorporate the user's profile attributes in the search and recommendation process

1. **Real-Time Data Integration**: Inclusion of real-time interaction with the user data to make dynamic recommendations.

2. **The methodology**: used in the field of modeling to date is called Advanced Modeling Techniques. The current methods used in the discipline of modeling to date are known as its Advanced Modeling Techniques. In-depth learning of the deep learning models and reinforcement learning models of CTR prediction.

3. User Profiling and Segmentation: Segmenting users according to their behavior to carry out a specific campaign.

4. Cross-Platform Personalization: Generalizing the recommendations across devices and platforms.

5. A/B Testing and Deployment: System performance evaluation in the real world.

14. Conclusion

The project demonstrates why behavioral based personalization should be crucial in online advertisements. Based on previous user data and context, we can fairly predict the way people will respond to ads. Taking a glance at the information and visualization, we receive certain positive ideas about what is liked by the users. It is also based upon the prediction model, which provides us with a foundation to make personal ad recommendations.
The paper demonstrates how data science can assist in addressing marketing issues, which will make the lives of users and advertisers easier. Despite few restrictions, the project has provided a framework, which can be applied to real ad platforms.

The obtained results have been manually verified and validated, and the performance and runtime of the algorithm have been evaluated using base line algorithms in different scenarios. The results show that our proposed algorithm can find potential users with higher accuracy than other models, especially when we need many target users. In the future, we will runthealgorithmonmoredatasetsandimproveitsefficiencytomakeitmoresuitablefor large datasets. Furthermore, we will incorporate the user's profile attributes in the search and recommendation process.

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